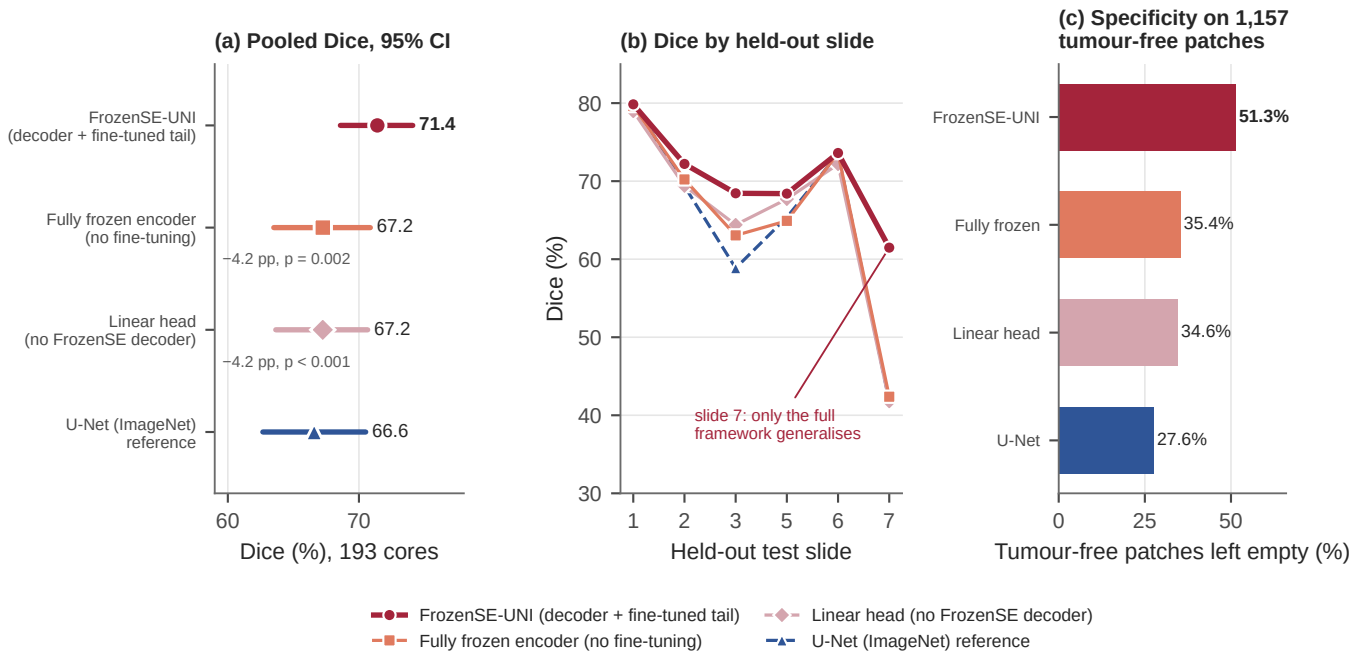
(a) Pooled Dice, 95% CI
FrozenSE-UNI (decoder + fine-tuned tail)
71.4
Fully frozen encoder (no fine-tuning)
67.2
−4.2 pp, p = 0.002
Linear head (no FrozenSE decoder)
67.2
−4.2 pp, p < 0.001
U-Net (ImageNet) reference
66.6
60
70
Dice (%), 193 cores
(b) Dice by held-out slide
Dice (%)
80
70
60
50
40
30
1
2
3
5
6
7
Held-out test slide
slide 7: only the full framework generalises
(c) Specificity on 1,157 tumour-free patches
FrozenSE-UNI
51.3%
Fully frozen
35.4%
Linear head
34.6%
U-Net
27.6%
0
25
50
Tumour-free patches left empty (%)
FrozenSE-UNI (decoder + fine-tuned tail)
Fully frozen encoder (no fine-tuning)
Linear head (no FrozenSE decoder)
U-Net (ImageNet) reference